

## Chapter 163 — Focal Therapy of Prostate Cancer — Numbers, Gold Standards and Traps

### IF YOU READ NOTHING ELSE

- **Focal therapy (FT)** = image-defined, biopsy-confirmed lesion ablation + **safety margin**.
- **Sweet spot: Gleason 3+4**, completely ablatable; **maximum 4+3** (2016 consensus).
- **Index lesion (IL)** = largest/highest-grade focus → drives natural history.
- **Stamey insignificant cancer**: <0.5 mL.
- **PROMIS** (multiparametric MRI, mpMRI): **sens 93%, spec 41%, positive predictive value (PPV) 51%, negative predictive value (NPV) 89%**.
- **Targeted + systematic biopsy** in every consensus; never mpMRI alone.
- **Margin**: 5 mm → >95%; 6 mm → 98.6% of significant cancer ablated.
- **Erectile dysfunction is NOT a contraindication** to FT.
- **Vascular-targeted photodynamic (VTP) therapy vs active surveillance (AS)**, randomised controlled trial (RCT): progression **28% vs 58%**; only level 1 FT evidence.
- **Outfield cancer**: short term = selection failure; long term = de novo.

### OPENING AND BIOLOGICAL BASIS

- **Incidence** 30.6/100,000/yr; **2nd most incident cancer** in men.
- **USA: commonest solid malignancy** in men; **2nd cancer death**.
- **Best mortality predictors**: grade, prostate-specific antigen (PSA), clinical stage; **grade earliest**.
- **Gleason 1973** = most practical prognosticator.
- **Hanahan–Weinberg hallmarks** (2000/2011): more in Gleason 4/5 than 3.
- **Pattern 4**: aneusomy 7, 8, 17 + HER-2/neu; **epidermal growth factor receptor (EGFR) copy number NOT raised**.
- **Deletions 6q, 16q** = commonest; size ∝ Gleason.
- **Multifocality 50–91%** at prostatectomy (table range 33–99%).
- **Unifocal 12.4–25%** → **52–93.4%** if satellites <0.5 mL discounted.
- **IL ≈ 80–90% of tumour volume**; source of most extraprostatic extension (EPE).
- **Aggressive feature in satellite, not IL: 12–38%**.
- **Modern significance thresholds**: >1.09 mL (Charing Cross); >1.31 mL (Rotterdam arm, European Randomised Study of Screening for Prostate Cancer).
- **First FT = hemiablation ("male lumpectomy")**.
- **Metastases monoclonal**; lethal clone once traced to small pattern 3 focus.
- **PIVOT**: prostatectomy survival benefit in **intermediate risk** only.
- **Strategy**: ablate grade 4 foci + **AS** for grade 3 remainder.

### PATIENT SELECTION

- **Selection = fundamental tenet**; patient must accept close surveillance.
- **Biopsy confirmation mandatory**; mpMRI alone insufficient.
- **Transrectal ultrasonography (TRUS) biopsy unilaterality**: high NPV, **poor specificity/PPV**; ceiling even at 24 cores.
- **Transperineal mapping biopsy (TMB)**: 5-mm grid; **>half of TRUS-unilateral disqualified**.
- **Unilateral predictors**: PSA ≤4 ng/mL + unilateral biopsy; **>2 positive cores** → more failure after FT.
- **Anterior cancers 15–33%**; missed by TRUS; **ideal FT targets** (both neurovascular bundles spared).
- **mpMRI for EPE**: spec 73–91%, sens 35–63%; **beats Partin tables**.
- **mpMRI best for >1 cm**, Gleason ≥7, index tumours.
- **mpMRI underestimates volume 5–30%**.
- **3 T > 1.5 T**; endorectal coil at 1.5 T; reader experience critical.
- **Prostate Imaging–Reporting and Data System (PI-RADS) v2 (2015)**: 5-point Likert; score ∝ Gleason.

- **In-bore biopsy**: most accurate, costly. **Cognitive fusion** = software fusion in experts.
- **Transperineal = transrectal MRI-targeted** detection of clinically significant cancer (csPCa).
- **~90% outfield csPCa within 1 cm** of target ("penumbra").
- **Prostate-specific membrane antigen (PSMA) 100–1000× overexpressed**; <sup>111</sup>In-capromab pendetide failed.
- **PSMA positron emission tomography (PET) vs mpMRI**: PET more specific, mpMRI more sensitive; PET unaffected by biopsy haemorrhage.
- **Shear-wave elastography cut-off 50 kPa**.

### Consensus cut-offs:

- **Residual 3+3**: ≤5 mm core (2013) → **≤1 mm (2016)**.
- **Max size (2016)**: 1.5 mL or 3 mL (not crossing midline); 20–25% gland (crossing).
- **Life expectancy (2013)**: >10 yr, not <5 yr; 2016: no limit.
- **2016**: ≥12-core systematic + MRI–TRUS fusion biopsy.
- **2019**: no cribriform/ductal; retreat to delay radical therapy.
- **Risk group**: intermediate.

### ABLATION PATTERNS AND TECHNOLOGIES

- **FT = lesion, quadrant, hemiablation**; hockey stick = partial ablation, not FT.
- **Transrectal → posterior**; **transperineal → anterior**.
- **Cryotherapy and irreversible electroporation (IRE)**: transperineal (metal probes, 17 G+); **high-intensity focused ultrasound (HIFU)** transrectal; laser either.

### Cryotherapy:

- **Joule–Thomson probes, 17 G**; variable ice length; visible ice edge; **urethral warmer**.
- **Thermocouples**: sphincter, neurovascular bundles, Denonvilliers fascia.
- **Biochemical progression-free survival (BPFS) 71–98%**; 3-yr 56–62.5%; 5-yr 62–70%.
- **Continence 95.1–100%**; **potency 46.8–100%** (half on phosphodiesterase type 5 inhibitors).
- **Clavien ≥3 ≤4%**; **retention commonest** → catheter a few weeks; **recto-urethral fistula (RUF) ≤0.8%**.
- **Cryo On-Line Data (COLD) registry**: partial = whole-gland BPFS; **better erections** (~2×, intermediate risk).

### HIFU:

- **≥55 °C**; posterior lesions; anterior hard (dissipation, oedema).
- **Limits**: size, height, calcification; **transurethral resection of prostate (TURP) beforehand** by some.
- **Infield 16.7–54%** (csPCa 0–21.8%); **outfield 7–42.8%** (csPCa 2–7.5%).
- **Continence 90–95%**; **potency 50–78%**.

- **HEAT registry (1379):** failure-free 68% intermediate, 65% high risk; **7-yr 69%.**
- **MRI-guided transurethral ultrasound ablation (TULSA):** MR thermometry; **precision 1.3 mm;** anterior gland reachable; grade 3 ~3%, no grade 4–5.

#### Laser:

- **≥60 °C;** ablation >5 mm beyond lesion; 14 G introducer, 17 G applicator.
- **Edge:** neovascularisation + mitoses → margin essential.
- **Infield 11–50%, outfield 16.7–60%;** continence 95–100%.
- **Laser RUF** healed with catheter in 4–6 weeks.

#### IRE:

- **Non-thermal;** probes 10–20 mm apart; >1600 V/cm.
- **20 test pulses → deep anaesthesia → 70 pulses/pair.**
- **csPCa infield 1–9.8%, outfield 5–16%.**
- **vs robot-assisted radical prostatectomy:** better continence and erections; some need salvage.

#### VTP:

- **TOOKAD (padeliporfin) 4 mg/kg IV; 753 nm,** 150 mW/cm, 200 J/cm.
- **RCT, 415 low risk:** progression 28% vs 58% AS.
- **RCT limits:** Gleason 6, short follow-up, surrogate end points, reclassification.
- **Perineal pain 15% VTP vs <1% AS.**
- **Radical therapy VTP vs AS:** 2 yr 7 vs 32%; 3 yr 15 vs 44%; 4 yr 24 vs 53%.

#### Other energies:

- **Radiofrequency ablation:** ~100 °C; ≤100 W; 450–1200 kHz; **fat limits conduction.**
- **High-dose-rate (HDR) brachytherapy:** stricture up to **8.2% in 2 yr** (apex, bulbar urethra).
- **Focal brachytherapy biopsy: 24 months.**
- **Focal HDR:** outfield csPCa 33%; new impotence 50%.
- **Stereotactic body radiotherapy tumour boost:** 5–10 fractions.
- **Urethral-sparing intensity-modulated radiotherapy:** no toxicity gain; **worse oncology.**
- **PRX302:** aerolysin activated by PSA.
- **Androgen deprivation + cryo:** no benefit.

#### Salvage (post-radiotherapy):

- **Cryo:** 5-yr BPFS ~50%; **RUF ~1.4%** = most feared.
- **HIFU:** 3-yr BPFS 48%; **continence 64–87.5%;** fistula ~2%; grade ≥3 ~quarter.
- **IRE:** progression-free 60%, metastasis-free 91%; continence 93%.
- **Focal = whole-gland salvage control; better function.**

## POST-FOCAL THERAPY FOLLOW-UP

- **PSA fall after FT:** 50–80%, variable.
- **Nadir + 1.0 ng/mL at 12 mo; nadir + 1.5 at 24–36 mo → biopsy.**
- **Biopsy:** mpMRI + **4–6 targeted cores at 3–6 mo;** mpMRI + systematic at **1–2 and 5 yr.**
- **Prostate Imaging after Focal Ablation (PI-FAB):** dynamic contrast-enhanced (DCE) read first; **enhancement >3 mm → biopsy.**
- **Success, infield:** short term eradication; long term no csPCa.
- **PI-FAB 1 monitor; 2 (DCE+) PSA kinetics; 3 (DCE+, diffusion-weighted imaging+) biopsy.**
- **PI-FAB 2 → 1** if linear, off original site; **2 → 3** if >3 mm or growing.
- **Contrast-enhanced ultrasound = mpMRI** for early ablation success; grey-scale TRUS cannot see zone.

## CONCLUSIONS

- **Whitmore:** "cure possible when necessary, necessary when possible?"
- **Up to half on AS** eventually have radical treatment.
- **No trial** of focal ablation + adjuvant whole-gland radiotherapy.

### CLASSIC TRAPS

- **Trap:** multifocal → no FT — **Actually:** IL + satellites <0.5 mL qualifies.
- **Trap:** MRI-negative → skip systematic biopsy — **Actually:** NPV 89%; biopsy still needed.
- **Trap:** MRI shows true size — **Actually:** underestimates 5–30%; add margin.
- **Trap:** impotence contraindicates FT — **Actually:** not a contraindication.
- **Trap:** transrectal HIFU for anterior tumour — **Actually:** posterior tool; anterior → transperineal/TULSA.
- **Trap:** PSA undetectable after FT — **Actually:** 50–80% fall; nadir + 1.0 at 12 mo.
- **Trap:** read T2 first after FT — **Actually:** PI-FAB reads DCE first.
- **Trap:** IRE spares bundles and urethra — **Actually:** fibrinoid necrosis, urethral denudation.
- **Trap:** VTP RCT shows survival gain — **Actually:** Gleason 6, surrogate end points.
- **Trap:** residual 3+3 tolerance loosened — **Actually:** tightened 5 mm → 1 mm.